

Tool: Entity-Relationship Diagram (ERD) + Access Pattern

Apa: blueprint data fitur — entity, relasi, kardinalitas, plus daftar pola akses (read/write) yang akan jalan ke skema. **Kapan dipake:** **sebelum** tulis DDL, tiap desain skema fitur baru (SOP-SH-01). ER dulu, baru tabel. **Kenapa:** desain dari access pattern (bukan tebak) = skema yang gak masalah 6 bulan lagi. Transparency (GCG) + dasar review joint dengan @saitama. **Output:** instance per fitur disimpan di `Log.md` / repo; keputusan Type-1 → Schema Decision Record di `adr/`.

TEMPLATE (copy mulai sini)

```
# ERD: <Nama Fitur>

**Owner:** Shikamaru · **Joint:** @saitama · **Status:** Draft / Approved / Locked
**Date:** YYYY-MM-DD · **Reversibility:** Type 1 / Type 2

## 1. Access patterns (audit DULU sebelum desain)
| Read/Write | Pattern | Frekuensi | Target latency |
|---|---|---|---|
| Read | ... | .../sec | < ...ms |
| Write | ... | .../sec | < ...ms |

## 2. Entity & relasi (kardinalitas)

- <Entity A> 1 — N <Entity B>    (A punya banyak B)
- <Entity B> N — 1 <Entity C>

## 3. Tabel (ringkas, DDL penuh di data-dictionary)
| Tabel | PK | Kolom kunci | FK | Audit cols? | deleted_at? |
|---|---|---|---|---|---|
| ... | ... | ... | ... |  |  |

## 4. Normalisasi & justifikasi

- Level: 3NF / denormalized (alasan: ...)

## 5. Catatan SACRED / PDP / Money

- Soft-delete: ...
- Data pribadi (PDP): ... → lihat data-classification-pdp
- Ledger uang isolated: ...
```

CONTOH TERISI (dari proyek contoh — fitur Payment + Order)

ERD: Proyek-Contoh — User, Order & Payment

Owner: Shikamaru · **Joint:** @saitama · **Status:** Locked
Date: 2026-05-29 · **Reversibility:** Type 1 (ada uang → escalate)

1. Access patterns

Read/Write	Pattern	Frekuensi	Target latency
Read	list order per user (paginated)	50/sec saat peak	< 30ms
Read	agregat total payment masuk per user	10/sec	< 50ms
Write	user submit order	20/sec spike	< 50ms
Write	catat payment masuk (uang)	5/sec	< 80ms

2. Entity & relasi

- users 1 — N orders (1 user banyak order)
- users 1 — N payments_ledger (1 user banyak transaksi payment)
- users 1 — N cart_item (item di keranjang)
- users 1 — N order_line

3. Tabel

Tabel	PK	Kolom kunci	FK	Audit cols?	deleted_at?
users	uuid	slug UNIQUE, owner_id	—	✓	✓
orders	uuid	user_id, nama_pembeli, hp, status(CHECK)	users	✓	✓
payments_ledger	uuid	user_id, amount, pengirim, masuk_at	users	✓ (created only)	✗
cart_item	uuid	user_id, item, harga, claimed_by	users	✓	✓
order_line	uuid	user_id, produk, is_active	users	✓	✓

4. Normalisasi & justifikasi

- 3NF semua. Tidak denormalize: total payment dihitung via agregat (atau materialized view kalau h

5. Catatan SACRED / PDP / Money

- Soft-delete: orders/cart/order_line pakai deleted_at. Payments_ledger TIDAK pernah dihapus/diubah
- Data pribadi (PDP): nama_pembeli, hp (orders); pengirim (payment) → diklasifikasi, lihat data-cl
- Ledger uang isolated: payments_ledger tabel sendiri, append-only, TIDAK campur profil owner/cust

Catatan: total payment **tidak** disimpan sebagai kolom redundan di **users** — itu update anomaly menunggu kejadian (uang ganda/hilang). Dihitung dari ledger. Kalau read jadi hot → materialized view, refresh on event. Itu cara aman buat data uang.